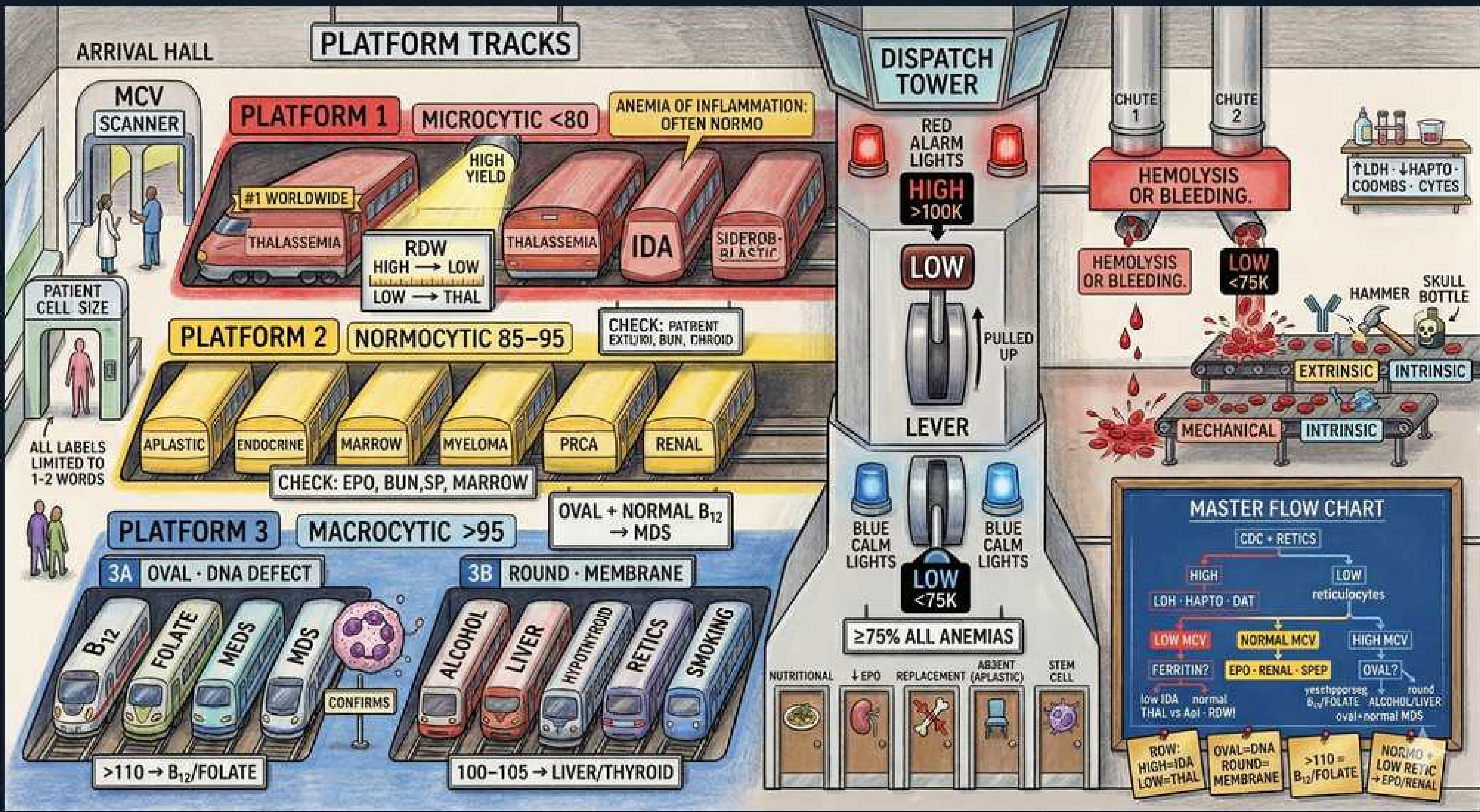
Anemia & CBC Interpretation — MCV Platforms



1. Arrival Hall — MCV scanner (patient cell size)

WHAT YOU SEE

Turnstile arrival hall with an 'MCV SCANNER' sizing each incoming patient cell.

WHAT IT ENCODES

Every CBC anemia is first triaged by mean corpuscular volume: the cell is 'scanned' for size and routed to a platform. Cutoffs: microcytic <80 fL, normocytic 80–100 fL, macrocytic >100 fL. MCV is the single most useful first branch point because it narrows the differential before any iron studies, B12, or marrow workup. Always pair MCV with the reticulocyte count to separate production from destruction.

BOARD TRAP

Picking a cause before scanning size is the trap — e.g. calling fatigue + low Hb 'iron deficiency' when MCV is normal. Normal MCV (normocytic) demands reticulocytes, not ferritin, as the next test.

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2. PLATFORM 1 — Microcytic <80

WHAT YOU SEE

Red 'PLATFORM 1 — MICROCYTIC <80' banner over a train of small red cars.

WHAT IT ENCODES

MCV <80 fL channels here. Mnemonic TAILS = Thalassemia, Anemia of chronic disease (often normo but can be micro), Iron deficiency, Lead, Sideroblastic. The dominant discriminators are iron studies and RDW. Hypochromia and target cells accompany the small cells on smear.

BOARD TRAP

Reflexively giving iron is the trap. Empiric iron in thalassemia trait causes iron overload — confirm low ferritin/low TSAT before treating, and never iron-load a normal-ferritin microcytic patient.

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3. #1 Worldwide / High-yield — Iron deficiency anemia

WHAT YOU SEE

Microcytic car tagged '#1 WORLDWIDE' and 'HIGH YIELD' (IDA).

WHAT IT ENCODES

IDA is the most common anemia worldwide and the highest-yield microcytic cause. Classic iron panel: low ferritin (most specific, <15–30 ng/mL), low serum iron, HIGH TIBC/transferrin, low transferrin saturation (<15–20%). RDW is HIGH (anisocytosis) and platelets are often reactively elevated. In adults, occult GI blood loss must be excluded — colonoscopy in men and postmenopausal women.

BOARD TRAP

Trap: ferritin is an acute-phase reactant, so a 'normal' ferritin in an inflamed patient can mask IDA — use TSAT and soluble transferrin receptor, or accept ferritin <100 as deficient when CRP is high.

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4. RDW high→low / Mentzer — Thalassemia vs IDA

WHAT YOU SEE

'RDW HIGH→LOW, LOW→HIGH→THAL' note beside the thalassemia car.

WHAT IT ENCODES

RDW separates the two big microcytic causes: HIGH RDW favors IDA (mixed cell sizes), NORMAL/LOW RDW favors thalassemia trait (uniformly small cells). Mentzer index = MCV ÷ RBC count; <13 suggests thalassemia (high RBC count, normal ferritin), >13 suggests IDA. Thalassemia shows normal/high ferritin, normal TIBC, target cells, and elevated HbA2 (beta-trait) on electrophoresis.

BOARD TRAP

Trap: ordering iron for a microcytic patient with a normal RDW and high-normal RBC count. That is thalassemia trait — iron is normal and supplementation is harmful.

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5. Sideroblastic anemia

WHAT YOU SEE

Microcytic car labeled 'SIDEROBLASTIC'.

WHAT IT ENCODES

Defective heme synthesis traps iron in mitochondria, producing ringed sideroblasts on marrow Prussian-blue stain. Iron studies paradoxically show HIGH ferritin and HIGH transferrin saturation despite anemia. Causes: lead poisoning, alcohol, isoniazid/B6 deficiency, copper deficiency/zinc excess, and myelodysplasia (often macrocytic). Pyridoxine (B6) may help hereditary X-linked forms.

BOARD TRAP

Trap: treating the high ferritin as iron overload needing iron, or as IDA. High ferritin + high TSAT in a microcytic anemia points to sideroblastic/lead, not iron deficiency.

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6. Anemia of inflammation (often normocytic)

WHAT YOU SEE

Top banner 'ANEMIA OF INFLAMMATION — OFTEN NORMO'.

WHAT IT ENCODES

Hepcidin-driven anemia of chronic disease is usually normocytic but can drift microcytic in chronic states. Pattern: LOW serum iron, LOW TIBC (the discriminator from IDA, where TIBC is high), normal-to-HIGH ferritin, low TSAT. It is a hypoproliferative anemia with low reticulocytes. Treat the underlying inflammation, not iron.

BOARD TRAP

Trap: confusing ACD with IDA because both have low serum iron. The TIBC splits them — HIGH TIBC = IDA, LOW/normal TIBC with high ferritin = anemia of inflammation.

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7. PLATFORM 2 — Normocytic 85–95

WHAT YOU SEE

Orange 'PLATFORM 2 — NORMOCYTIC 85–95' banner with a train of normal-size cars.

WHAT IT ENCODES

MCV 80–100 fL routes here, then is split by reticulocyte count. LOW retics = hypoproliferative (marrow or EPO problem): aplastic anemia, marrow infiltration/myeloma, endocrine, renal failure (low EPO), pure red cell aplasia. HIGH retics = hemolysis or acute blood loss. Corrected retic = retic% × (Hct/45); reticulocyte production index >2 means adequate marrow response.

BOARD TRAP

Trap: skipping the reticulocyte count. Without it you cannot tell a marrow that is failing (low retic) from one compensating for hemolysis/bleeding (high retic) — the entire normocytic workup pivots on this number.

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8. Renal failure / PRCA — check EPO, BUN

WHAT YOU SEE

Normocytic cars 'RENAL' and 'PRCA' with 'CHECK: EPO, BUN, SP, MARROW'.

WHAT IT ENCODES

Chronic kidney disease causes a hypoproliferative normocytic anemia from inadequate erythropoietin — confirm with elevated BUN/creatinine and low/inappropriately-normal EPO, treat with ESAs once iron-replete. Pure red cell aplasia is isolated absent reticulocytes/erythroid precursors, classically from parvovirus B19 (especially in sickle cell aplastic crisis), thymoma, or anti-EPO antibodies. Workup: EPO level, BUN, serum protein/SPEP (myeloma), bone marrow.

BOARD TRAP

Trap: giving ESAs to a CKD patient who is iron-deficient — ESAs fail without adequate iron stores. Always check ferritin/TSAT and target TSAT >20% before or alongside ESA therapy.

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9. Multiple myeloma / marrow infiltration

WHAT YOU SEE

Normocytic cars 'MYELOMA' and 'MARROW'.

WHAT IT ENCODES

Marrow infiltration (myeloma, leukemia, metastasis, fibrosis) crowds out erythropoiesis, giving normocytic anemia with low retics and often a leukoerythroblastic smear (teardrop cells, nucleated RBCs, left-shifted granulocytes). Myeloma clues: rouleaux, high total protein/low albumin gap, hypercalcemia, renal failure, lytic bone lesions; confirm with SPEP/UPEP, free light chains, and marrow. Check serum protein (SP) as the scene notes.

BOARD TRAP

Trap: attributing an older patient's normocytic anemia + back pain + high calcium + renal failure to 'aging' — that CRAB picture is myeloma until SPEP/light chains prove otherwise.

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10. PLATFORM 3 — Macrocytic >95

WHAT YOU SEE

Blue 'PLATFORM 3 — MACROCYTIC >95' banner with a train of large blue cars.

WHAT IT ENCODES

MCV >100 fL routes here and splits into 3A megaloblastic (oval macrocytes, DNA-synthesis defect) and 3B non-megaloblastic (round macrocytes, membrane/lipid defect). The smear is the key fork: hypersegmented neutrophils (>5 lobes) = megaloblastic. Very high MCV >110–115 strongly favors B12/folate deficiency over the milder macrocytosis of liver disease or alcohol.

BOARD TRAP

Trap: calling all macrocytosis B12/folate. Round macrocytes without hypersegmentation point to alcohol/liver/hypothyroid/reticulocytosis — check the smear and B12/folate before treating.

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11. 3A — Oval / DNA defect: B12 & Folate

WHAT YOU SEE

Box '3A OVAL — DNA DEFECT' over B12 and FOLATE cars; '>110 → B12/FOLATE'.

WHAT IT ENCODES

Megaloblastic anemia from impaired DNA synthesis: B12 deficiency, folate deficiency, methotrexate/antimetabolites, MDS. Discriminate with metabolites — both B12 and folate deficiency raise homocysteine, but only B12 deficiency raises methylmalonic acid (MMA). B12 deficiency adds neurologic signs (subacute combined degeneration, paresthesias) that folate deficiency lacks. ALWAYS replace B12 before folate — giving folate alone corrects the anemia but lets neurologic damage progress irreversibly.

BOARD TRAP

Trap: giving folate to a B12-deficient patient and watching the CBC normalize while myelopathy worsens. Check/replace B12 first; an elevated MMA is the discriminator (high in B12, normal in pure folate deficiency).

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12. MDS — confirms macrocytosis

WHAT YOU SEE

'MDS' car marked 'CONFIRMS' in platform 3A.

WHAT IT ENCODES

Myelodysplastic syndrome is a clonal marrow disorder of older adults causing macrocytic anemia with cytopenias and dysplastic cells (e.g. pseudo-Pelger-Huet neutrophils, ringed sideroblasts). It is megaloblast-mimicking but B12/folate are normal — diagnosis is by marrow with dysplasia and cytogenetics. Risk stratify with IPSS-R; del(5q) responds to lenalidomide.

BOARD TRAP

Trap: chasing B12/folate repeatedly in an older patient with refractory macrocytic anemia and normal vitamins. Persistent unexplained macrocytic cytopenias warrant a bone marrow biopsy for MDS.

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13. 3B — Round / membrane: alcohol, liver, hypothyroid

WHAT YOU SEE

Box '3B ROUND — MEMBRANE' over ALCOHOL, LIVER, HYPOTHYROID, RETICS, SMOKING; '100–105 → LIVER/THYROID'.

WHAT IT ENCODES

Non-megaloblastic macrocytosis: round macrocytes, NO hypersegmented neutrophils, normal B12/folate. Causes are membrane-cholesterol loading and marrow stress — alcohol (direct toxicity), chronic liver disease (target cells, lipid changes), hypothyroidism, reticulocytosis (large young cells), and smoking. MCV is usually only mildly elevated (100–110); the 100–105 range especially points to liver or thyroid.

BOARD TRAP

Trap: ignoring reticulocytosis as a macrocytosis cause — a hemolyzing or recovering patient has many large reticulocytes that raise MCV. Don't work this up as B12 deficiency; the retic count and absent hypersegmentation give it away.

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14. Dispatch Tower — platelet alarms (HIGH >100K / LOW <75K)

WHAT YOU SEE

Control tower with RED ALARM lights 'HIGH >100K', 'LOW', and BLUE CALM lights '>75K'.

WHAT IT ENCODES

The platelet tower monitors thrombocytopenia thresholds. Spontaneous bleeding risk rises sharply below ~10–20K (consider prophylactic transfusion at <10K). Most procedures need >50K, and neurosurgery/epidural ~>80–100K. Evaluate isolated thrombocytopenia by smear first — clumping (pseudothrombocytopenia), schistocytes (TMA), blasts (leukemia) — before assuming ITP.

BOARD TRAP

Trap: transfusing platelets in TTP or HIT — it can worsen thrombosis. In a thrombocytopenic patient with schistocytes (TTP) or recent heparin (HIT), platelet transfusion is contraindicated; treat the underlying process.

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15. Hemolysis OR Bleeding branch — high retics

WHAT YOU SEE

Red 'HEMOLYSIS OR BLEEDING' banner with a 'LOW <75K' chute and a 'pulled-up' lever.

WHAT IT ENCODES

High reticulocytes route here. Hemolysis labs: HIGH LDH and HIGH indirect (unconjugated) bilirubin, LOW haptoglobin, high retics. Intravascular hemolysis adds hemoglobinemia/hemoglobinuria and very low haptoglobin (MAHA, PNH, transfusion reactions, mechanical valves). Extravascular (spherocytosis, warm AIHA, hypersplenism) shows splenic clearance with less hemoglobinuria.

BOARD TRAP

Trap: calling jaundice + anemia 'liver disease.' Indirect hyperbilirubinemia with high LDH, low haptoglobin, and high retics is hemolysis — check the DAT/Coombs and smear, not just LFTs.

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16. Coombs / DAT — extrinsic vs intrinsic, mechanical

WHAT YOU SEE

'EXTRINSIC / INTRINSIC' and 'MECHANICAL / INTRINSIC' boxes with a hammer and skull bottle (LDH/hapto/bili, Coombs).

WHAT IT ENCODES

The DAT (direct antiglobulin/Coombs) splits immune from non-immune hemolysis. DAT-positive = autoimmune hemolytic anemia: warm IgG (lymphoma, SLE, drugs; treat steroids) vs cold IgM (Mycoplasma, EBV, cold agglutinins; avoid cold, rituximab). DAT-negative with schistocytes = mechanical/microangiopathic (TTP/HUS, DIC, prosthetic valve, malignant HTN). Intrinsic causes (membrane, enzyme G6PD, hemoglobinopathy) are Coombs-negative and often inherited.

BOARD TRAP

Trap: starting steroids for every hemolysis. A DAT-negative hemolytic anemia with schistocytes is microangiopathic — steroids won't help; you need plasma exchange (TTP) or to treat DIC/valve, and platelet transfusion is harmful in TTP.

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17. Master Flow Chart — CBC → retics → branches

WHAT YOU SEE

'MASTER FLOW CHART' panel: CBC→retics, low MCV / normo / high, hemolysis workup (LDH, hapto, DAT).

WHAT IT ENCODES

The integrating algorithm: start with CBC + reticulocyte count. Branch by MCV (low → iron studies/TAILS; normal → retic-driven split; high → smear for hypersegmentation + B12/folate). For any high-retic anemia, run the hemolysis panel (LDH, haptoglobin, indirect bilirubin, DAT) and a smear. The retic count threads through every arm — it is the second universal test after MCV.

BOARD TRAP

Trap: ordering a shotgun panel instead of following the tree. The boards reward a sequenced workup — MCV then reticulocytes then targeted tests — over reflexively sending B12, iron, and Coombs all at once.
